Requirements Table — Problem #08: Digital Campus Library Reservation Gateway

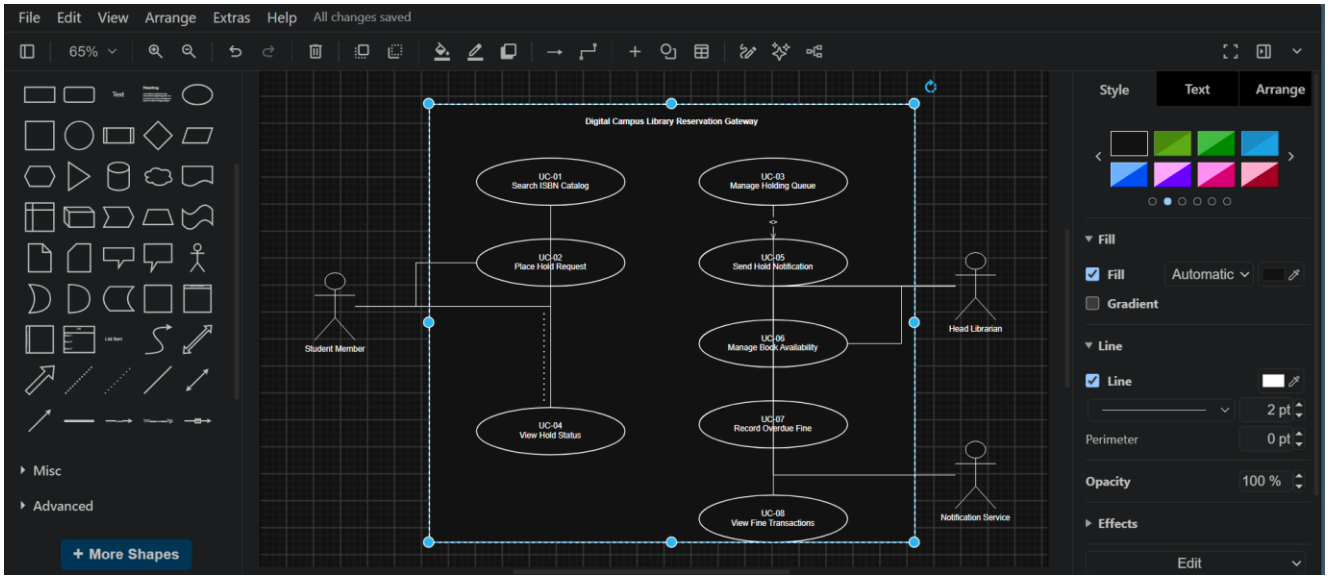
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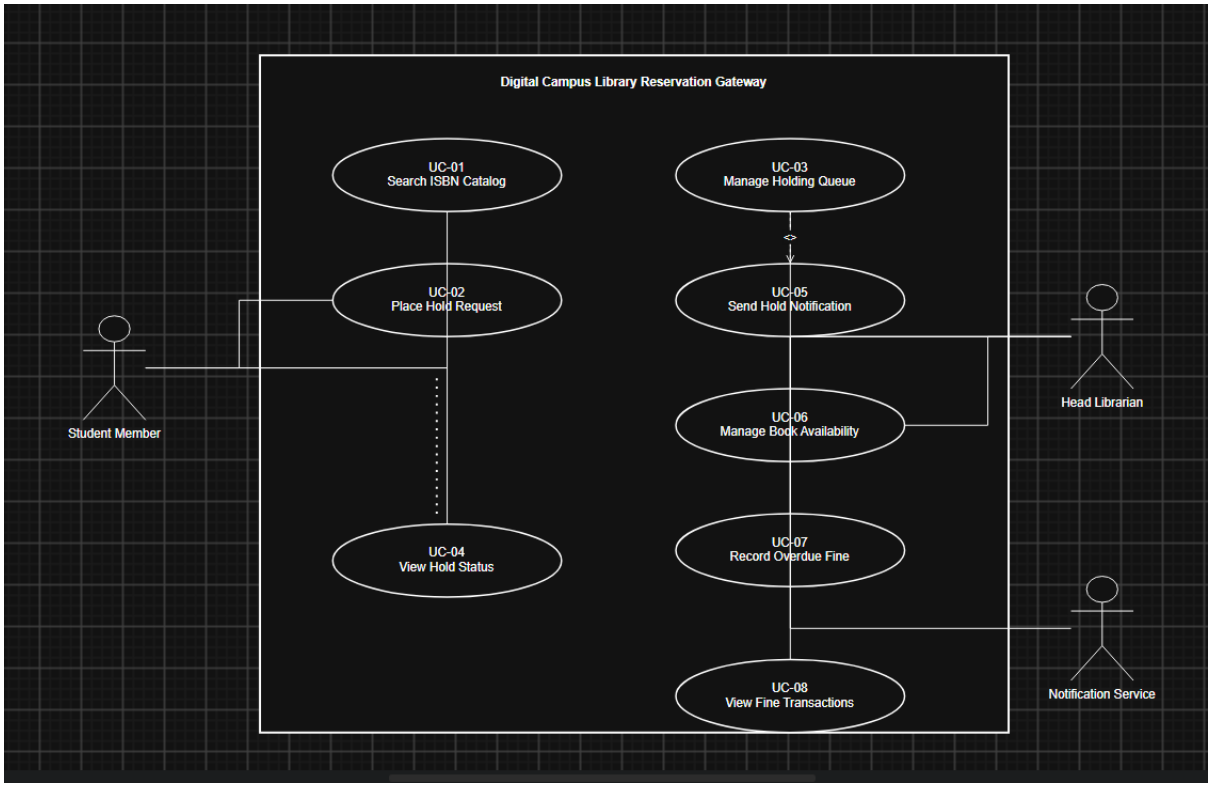
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow Student Members to search the ISBN repository and place hold requests on currently loaned books by entering the FIFO waitlist.	High	**Pass:** For a valid ISBN of a currently loaned book, the system creates a hold request and displays the student's FIFO queue position. **Fail:** The request is not created or no queue position is assigned.	Core functionality specified in the problem statement; enables students to reserve borrowed books fairly.
FR-002	Functional	The system shall maintain a FIFO holding queue for each book and assign each new valid hold request the next available queue position.	High	**Pass:** If three students already occupy positions 1–3, the next valid request receives position 4. **Fail:** A later request receives a position before an earlier valid request.	Ensures reservations are processed fairly according to request order.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-003	Functional	The system shall automatically notify the next Student Member in the holding queue when a reserved book becomes available for collection.	High	**Pass:** When a held copy becomes available, the system sends a notification to the Student Member at the front of the queue and records the notification. **Fail:** The eligible student receives no notification.	The problem statement specifically requires automated holding notifications.
FR-004	Functional	The system shall allow the Head Librarian to view, update, and remove hold requests and holding-queue entries for library books.	Medium	**Pass:** The Head Librarian can select a book and successfully view its queue, update an entry, or remove an entry, with the change reflected immediately. **Fail:** Any authorized queue-management operation	Gives the librarian administrative control over reservation queues.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
				cannot be completed.	
FR-005	Functional	The system shall record overdue fine transactions associated with Student Member borrowing records and allow the Head Librarian to view the recorded transactions.	Medium	**Pass:** When an overdue fine transaction is recorded, the transaction contains the student identifier, amount, date, and status and is visible to the Head Librarian. **Fail:** Any required transaction information is missing or the transaction cannot be retrieved.	The scenario explicitly includes tracking overdue fine transactions.
NFR-001	Nonfunctional — Performance & Security	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	**Pass:** Benchmarking tests show catalog search results are returned in **<150 ms** for a 50,000-title repository	This is the performance requirement explicitly provided in the problem statement.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
				under the specified test conditions. **Fail:** The measured response time is **≥150 ms**.	
NFR-002	Nonfunctional — Availability	The system shall preserve successfully submitted hold requests and their FIFO queue positions without loss during normal system operation.	High	**Pass:** After a successful hold submission, restarting or refreshing the application does not remove the request or change its stored queue position. **Fail:** A successfully submitted request or its queue position is lost or changed.	Prevents loss of reservation information and protects the integrity of the FIFO queue.


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Digital Campus Library Reservation Gateway
UC-01 Search ISBN Catalog
UC-02 Place Hold Request
UC-03 Manage Holding Queue
UC-04 View Hold Status
UC-05 Send Hold Notification
UC-06 Manage Book Availability
UC-07 Record Overdue Fine
UC-08 View Fine Transactions
Student Member
Head Librarian
Notification Service
Style
Text
Arrange
Fill
Automatic
Gradient
Line
2 pt
Perimeter
0 pt
Opacity
100 %
Effects
Edit
Misc
Advanced
+ More Shapes


Digital Campus Library Reservation Gateway
UC-01 Search ISBN Catalog
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UC-08 View Fine Transactions
Student Member
Head Librarian
Notification Service

Main Success Scenario

1. **Student Member** logs into the Digital Campus Library Reservation Gateway.
2. The system authenticates the Student Member and displays the library search interface.
3. The Student Member enters the **ISBN** or searches for the required book.
4. The system searches the ISBN repository and displays the matching book details.
5. The Student Member selects a book that is currently loaned.
6. The system displays the book's current availability/holding information and provides the **Place Hold** option.
7. The Student Member selects **Place Hold**.
8. The system validates the Student Member's eligibility to place the hold.
9. The system creates a new hold request for the selected book.
10. The system places the request at the **end of the FIFO holding queue**.
11. The system assigns the next queue position to the Student Member.
12. The system displays a confirmation containing the book details and the Student Member's queue position.
13. The use case ends successfully.

This follows the core requirement supplied in the assignment: members search the ISBN repository and place holds on currently loaned books, with the request entering a FIFO waitlist and receiving the correct queue position.

Alternate Flows

A1. Book Is Not Currently Loaned

1. At Step 5, the Student Member selects a book that is currently available.

2. The system does not create a hold request.
3. The system displays that the book is currently available and provides the normal borrowing/availability option.
4. The use case ends without creating a hold.

A2. Student Member Is Not Eligible for Another Hold

1. At Step 8, the system determines that the Student Member has reached the maximum permitted number of active holds/borrowed books.
2. The system rejects the hold request.
3. The system displays an explanatory message.
4. No queue position is assigned.
5. The use case ends.

This is consistent with the assignment's supplied acceptance example, which states that a member who is already holding the maximum allowed books must not be able to place an additional reservation.

A3. Duplicate Hold Request

1. At Step 8, the system detects that the Student Member already has an active hold for the same book.
2. The system does not create a duplicate request.
3. The system displays the existing hold/queue position to the Student Member.
4. The use case ends.

Result

Successful outcome: The Student Member is added to the correct FIFO queue position and receives confirmation of the reservation.